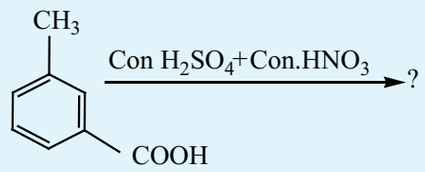
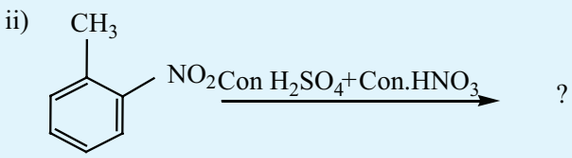
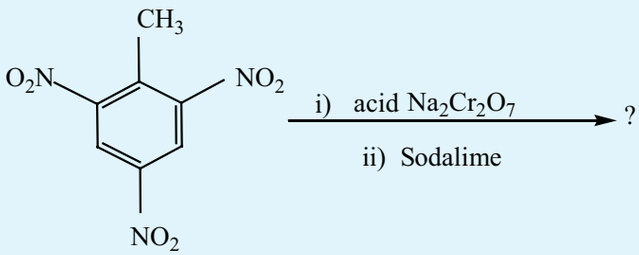
Nitrobenzene does not undergo Friedel – Crafts reactions due to the strong deactivating nature of $-NO_2$ group.

**Evaluate yourself** Predict the major product that would be obtained on nitration of the following compounds

i) 


ii) 


iii) 


## 13.2 Amines - clasification

Amines

- Aliphatic amines
- Aromatic amines

| | | Aliphatic amines | Aromatic amines |
|---|---|---|---|
| Primary | | $CH_3-CH_2-NH_2$ ethanamine | $C_6H_5-NH_2$ benzenamine(aniline) |
| Secondary | simple | $CH_3-NH-CH_3$ N-methylmethanamine | $C_6H_5-NH-C_6H_5$ N-phenylbenzenamine |
| | mixed | $CH_3-NH-CH_2-CH_3$ N-methylethanamine | $C_6H_5-NH-CH_3$ N-phenylmethanamine |
| Tertiary | simple | $CH_3-N(CH_3)-CH_3$ N,N-dimethylmethanamine | $C_6H_5-N(C_6H_5)-C_6H_5$ N,N-diphenylbenzenamine |
| | mixed | $CH_3-N(CH(CH_3)_2)-CH_2-CH_3$ (N-ethyl-N-methyl) propan–2–amine | $C_6H_5-N(CH_3)-CH_2-CH_3$ N-methyl-N-phenylethanamine |



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